llmXive Automated Review of A Survey of Large Audio Language Models: Generalization, Trustworthiness, and Outlook

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript presents a comprehensive survey of Large Audio Language Models (LALMs), but several specific factual claims regarding benchmark results and statistical findings require verification against their cited sources to ensure accuracy.

In Section 5.1.3 ("From Fact Retrieval to Personal Alignment"), the text states that MMSU [wang2025mmsu] reveals models achieve "only 53.60% and 39.35% accuracy on phonology and paralinguistics." While the numbers 53.60 and 39.35 appear in the critical elements list, the text does not explicitly confirm that these specific values correspond to the phonology and paralinguistics sub-tasks respectively in the source paper. There is a risk of conflation with other metrics or benchmarks. The authors must verify that these exact figures are attributed to the correct sub-tasks in the MMSU paper.

In Section 5.3.1 ("Jailbreaks and Acoustic Backdoors"), the claim that JALMBench [peng2025jalmbench] shows audio attacks achieve higher success rates (21.5%) than text (17.0%) is a strong comparative statement. The text summary includes the numbers 21.5 and 17.0, but the context of the comparison (e.g., specific model, specific attack type, or dataset subset) is not fully detailed in the prose. It is crucial to confirm that the source paper explicitly compares audio and text attack success rates under identical conditions to support this direct comparison.

In Section 5.2.1 ("Temporal Robustness in Long-Form Context"), the manuscript claims ChronosAudio [luo2026chronosaudio] reveals "performance drops exceeding 90%." The phrasing "exceeding 90%" implies a catastrophic failure (remaining accuracy <10%). The critical element list includes "90", but without the full source text, it is unclear if the paper reports a 90% drop or if the metric is something else (e.g., 90% of instances failed). The authors should clarify the exact metric and ensure the phrasing "exceeding 90%" is mathematically precise based on the source data.

Finally, in Section 5.3.2 ("Privacy, Fairness, and Authentication"), the claim that AudioTrust [li2025audiotrust] reveals ">35% bypass rates for voice cloning" relies on the number 35 found in the critical elements. The term "bypass rate" can vary in definition (e.g., success rate of a specific attack vs. overall system failure). The authors should verify that the source paper defines "bypass rate" in a way that aligns with the claim and that the 35% figure is a robust aggregate statistic rather than an outlier.

These issues are primarily writing-level concerns regarding the precision of claim attribution, but they are critical for the scientific accuracy of a survey paper. Correcting these citations and ensuring the numbers match the source papers exactly is necessary before acceptance.

Action items.

- **[writing]** Section 5.1.3 claims MMSU [wang2025mmsu] reveals 53.60% and 39.35% accuracy on phonology and paralinguistics. The citation key 'wang2025mmsu' appears in the bibliography but the specific numeric breakdown is not verifiable in the provided text summary. Verify these exact figures against the source paper to ensure they are not conflated with other benchmarks.

- [**writing**] Section 5.3.1 states JALMBench [peng2025jalmbench] shows audio attacks achieve 21.5% success vs 17.0% for text. The text summary lists '21.5' and '17.0' as critical elements but does not explicitly link them to this specific comparison in the source text. Confirm the source paper supports this specific head-to-head comparison and that the numbers are not from different experimental settings.
- [**writing**] Section 5.2.1 claims ChronosAudio [luo2026chronosaudio] reveals performance drops exceeding 90% in long contexts. The text summary lists '90' as a critical element. Verify if the source paper reports a 'drop exceeding 90%' (i.e., remaining accuracy <10%) or if the metric is different (e.g., 90% of tasks failed). Ensure the phrasing 'exceeding 90%' accurately reflects the data.
- [**writing**] Section 5.3.2 claims AudioTrust [li2025audiotrust] reveals open-source models have >35% bypass rates for voice cloning. The text summary lists '35' as a critical element. Verify the specific metric (bypass rate) and the threshold (35%) in the source paper, as 'bypass rate' can be defined differently across security benchmarks.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-structured conceptual taxonomy that effectively visualizes the three pillars of trustworthy LALM evaluation described in the caption. The diagram is uncluttered, the text is legible, and the specific failure scenarios listed in the sub-boxes align perfectly with the definitions provided in the caption.

Figure 2

The figure clearly illustrates the Audio-CoT workflow steps, but it fails to support the caption's claim of a comparative analysis because the standard LALM model is not visualized for comparison.

Figure 3

The figure effectively illustrates the six dimensions of trustworthiness with clear visual examples, but contains minor typographical errors in the text labels (‘Stanard’ and ‘Mr.John’) that should be corrected.

Figure 4

Figure 4 attempts to show cumulative growth but suffers from a confusing x-axis timeline, a grammatical error in the y-axis label, and potential misalignment between plotted data points and the y-axis scale, undermining its clarity and accuracy.

Figure 5

The figure provides a clear and readable conceptual taxonomy of LALM evaluation pillars. However, the caption is an exact duplicate of Figure 1’s text, and the title font style is stylistically inconsistent with the rest of the document.

Figure 6

The figure clearly illustrates the Audio-CoT workflow but fails to deliver on the caption’s promise of a comparative analysis, as the standard direct-response model is not visualized.

Figure 7

Figure 7 effectively illustrates the six dimensions of LALM trustworthiness with clear visual examples, but contains minor typographical and grammatical errors in the text labels (‘Stanard’, ‘Mr.John’) that should be corrected.

Figure 8

The figure effectively visualizes the growth of trustworthy LALM research with clear categorization, but the caption contains a grammatical error (‘almost scholarly’) and the timeline projects into the future, which is inconsistent with a retrospective survey.

Action items.

- [**science**] Figure 2: The caption claims a ‘comparative analysis’ between ‘standard LALM’ and ‘Audio-CoT’, but the figure only visualizes the Audio-CoT workflow. The ‘standard direct-response’ model mentioned in the caption is not shown, making the comparison impossible to verify visually.
- [**writing**] Figure 2: The caption contains a missing space after the period in ‘Audio-CoT.This’, which affects readability.
- [**writing**] Figure 3: The label ‘Stanard/Dialect’ in the Fairness example box contains a typo and should be ‘Standard/Dialect’.
- [**writing**] Figure 3: The ‘Hallucination’ example box contains a grammatical error (‘Mr.John’

should be ‘Mr. John’ with a space).

- [**science**] Figure 4: The x-axis timeline is logically inconsistent; the ‘2025.1-3’ period is placed to the right of ‘2024-before’ but to the left of ‘2025.4-6’, yet the ‘2025.1-3’ label is visually positioned after ‘2024-before’ while the ‘2025.4-6’ label is further right, creating a confusing non-linear or broken sequence that contradicts the caption’s claim of tracking growth over time.
- [**writing**] Figure 4: The y-axis label ‘Number of Papers(cumulated)’ contains a grammatical error; ‘cumulated’ should be ‘cumulative’ to match standard scientific terminology and the caption’s phrasing.
- [**science**] Figure 4: The data points (yellow diamonds) and their corresponding numerical labels (e.g., ‘1+’, ‘4+’, ‘10+’) do not align with the y-axis scale; for instance, the ‘10+’ point is plotted near y=10, but the ‘13+’ point is plotted near y=13, while the ‘12+’ point is plotted near y=12, suggesting the y-axis values are not accurately representing the cumulative count as implied by the axis label.
- [**writing**] Figure 5: The caption is a verbatim duplicate of Figure 1’s caption, suggesting a copy-paste error in the manuscript preparation.
- [**writing**] Figure 5: The title ‘Trustworthy LALM Evaluation’ is rendered in a casual, handwritten-style font that is inconsistent with the professional tone of a scientific survey.
- [**science**] Figure 6: The caption claims a ‘comparative analysis’ between standard LALMs and Audio-CoT, but the figure only visualizes the Audio-CoT workflow. It lacks the ‘standard direct-response’ model side to support the comparison.
- [**writing**] Figure 6: The caption contains a missing space after the period in ‘Audio-CoT.This’, which affects readability.
- [**writing**] Figure 7: The label ‘Stanard/Dialect’ in the Fairness example box contains a typo and should be ‘Standard/Dialect’.
- [**writing**] Figure 7: The ‘Hallucination’ example box contains a grammatical error (‘Mr.John’ should be ‘Mr. John’ with a space).
- [**writing**] Figure 8: The caption contains a grammatical error (‘surge in almost scholarly publications’) which likely should read ‘a surge in scholarly publications’ or ‘almost 20 publications’.
- [**science**] Figure 8: The x-axis timeline projects into the future (2025.4-6 through 2026.1-3), implying the chart predicts or includes future data points rather than tracking historical research, which contradicts the caption’s claim to ‘track’ existing publications.
- [**writing**] Figure 8: The y-axis label ‘Number of Papers(cumulated)’ uses non-standard terminology; ‘Cumulative Count’ or ‘Cumulative Number’ is standard.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript suffers from significant jargon overuse, frequently employing specialized terminology from signal processing, economics, and deep learning without definition or simplification. This creates a barrier for non-specialist readers, violating the goal of a comprehensive survey.

In the **Introduction** and **Section 1**, terms like “non-stationary auditory signals,” “structured semantic latent spaces,” and “Endogenous Mechanisms” are used without explanation. “Endogenous” is particularly unnecessary; “Internal” conveys the same meaning more clearly. Similarly, “latent spaces” should be defined as “hidden representations” or “internal data structures” for clarity.

Section 2 introduces dense jargon such as “modality nexus,” “factorized tokenization,” and “Pareto frontier.” “Nexus” is a flowery synonym for “connection” that adds no precision. “Pareto frontier” is an economic concept that should be replaced with “optimal trade-off boundary” or “best possible balance” to ensure accessibility.

Section 5 (Evaluation) and **Section 4** (Safety) continue this trend with phrases like “structural attention dilution,” “plasticity-stability dilemma,” “intrinsic representation engineering,” and “reasoning tax–shield bifurcation.” These compound metaphors and technical terms obscure the underlying concepts. For instance, “plasticity-stability dilemma” is a known cognitive science term but should be explicitly defined or paraphrased as the conflict between learning new information and retaining old knowledge. “Reasoning tax–shield bifurcation” is almost unintelligible without a glossary; it should be simplified to “the trade-off where security measures reduce reasoning ability.”

Finally, specific signal processing terms like “Mel-frequency bins” (Section 4) and “disentanglement efficiency” (Section 5.1.1) appear without context. While some technical depth is expected, a survey paper must define these terms upon first use or use plainer alternatives to remain inclusive. The current density of undefined acronyms (e.g., LALM, ASR, RLHF) and specialized vocabulary significantly hinders readability for the intended broad audience.

Action items.

- [**science**] The manuscript suffers from significant jargon overuse, frequently employing specialized terminology from signal processing, economics, and deep learning without definition or simplification. This creates a barrier for non-specialist readers, violating the goal of a comprehensive survey. In the Introduction and Section 1, terms like “non-stationary auditory signals,” “structured semantic latent spaces,” and “Endogenous Mechanisms” are used without explanation. “Endogenous” is particularly unnece

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript generally maintains a coherent narrative structure, moving from architectural foundations to trustworthiness challenges and future outlooks. However, several logical gaps exist where conclusions are asserted without sufficient mechanistic support or where premises do not fully entail the stated conclusions.

First, the central claim in the Conclusion regarding the "asymmetry" between mature offensive research and "rudimentary" defenses requires tighter logical support. While Section 5.3.1 and 5.3.2 enumerate several defense strategies (e.g., ALMGuard, SARSteer, AudioSafe) and benchmarks, the text asserts these are "rudimentary" without explicitly defining the criteria for this judgment. Is the claim that these defenses are ineffective, or that they are reactive rather than proactive? The current text lists the existence of defenses, which logically contradicts the claim that they are "rudimentary" unless the *efficacy* is the specific metric being critiqued. The argument would be stronger if it explicitly linked the listed defenses to their failure modes or limited scope, rather than just listing them and then declaring the field immature.

Second, in Section 5.2.1, the paper cites ChronosAudio to claim a >90% performance drop in long contexts due to "Structural Attention Dilution." The term "Structural Attention Dilution" is introduced as the causal mechanism but is never defined or explained in the text. The logical chain is broken here: the evidence (performance drop) is presented, and a label (attention dilution) is attached, but the mechanism connecting the two is missing. Without explaining *how* the structure causes the dilution and *why* that leads to such a drastic drop, the causal claim remains an unsupported assertion.

Finally, the proposal in Section 5.4 for "Causal Auditory World Modeling" as a solution to hallucination lacks a direct logical bridge. The paper defines hallucination as a failure to ground outputs in acoustic reality (Sec 5.1). While world modeling implies a deeper understanding of physical dynamics, the text does not explain why this specific capability is the necessary solution to grounding failures, as opposed to, for example, improved attention mechanisms or better training data. The leap from "models hallucinate" to "we need causal world modeling" is a gap in the argumentation that needs to be filled with a specific rationale.

Action items.

- **[science]** The conclusion claims a 'developmental asymmetry' where offensive research is mature and defenses are rudimentary. However, Section 5.3.1 and 5.3.2 list numerous specific defense mechanisms (ALMGuard, SARSteer, AudioSafe) and benchmarks. The text must clarify whether the 'rudimentary' claim refers to the *effectiveness* of these defenses rather than their *existence*, or provide evidence that these defenses fail in practice to support the causal claim.
- **[writing]** Section 5.2.1 cites ChronosAudio showing >90% performance drop due to 'Struc-

tural Attention Dilution'. The paper does not define this mechanism or explain the causal link between attention dilution and the specific magnitude of the drop. The argument relies on an undefined term to explain a quantitative result, breaking the logical chain.

- **[science]** The 'Future Outlook' (Sec 5.4) proposes 'Causal Auditory World Modeling' as a solution to hallucination. The paper defines hallucination as a failure of acoustic grounding (Sec 5.1) but does not logically demonstrate why 'world modeling' (a counterfactual reasoning capability) is the necessary or sufficient condition to fix grounding failures, rather than just better alignment.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The manuscript presents a comprehensive survey of Large Audio Language Models (LALMs), but several sections exhibit over-claiming by extrapolating specific benchmark results to universal properties of the field or by using absolute language that the cited evidence does not fully support.

First, the Conclusion and Section 5.4 assert a stark asymmetry: "offensive research is mature, while defenses are rudimentary." This is an over-generalization. The paper itself dedicates significant space to detailing specific, quantitative defense mechanisms such as ALMGuard, SARSteer, and AudioSafe, which demonstrate non-trivial success rates. Describing these as "rudimentary" dismisses the nuance of the cited work. The authors should rephrase this to indicate that defenses are currently "fragmented," "less standardized," or "reactive" rather than fundamentally immature, to accurately reflect the evidence provided in their own survey.

Second, in Section 5.3.2, the text states that "HearSay shows models infer gender with >92% accuracy." This phrasing implies a universal capability of the LALM class. However, this statistic is derived from a specific evaluation on a specific dataset (HearSay). Extrapolating this to a general property of all LALMs without qualifying that it applies to models evaluated on that specific benchmark or under those specific conditions is an overreach. The claim should be scoped to "models evaluated on HearSay" or similar to maintain scientific rigor.

Finally, Section 5.1.1 introduces terms like "Perception-Cognition Gap" and "Denoising Paradox" as if they are established, named phenomena governing the field. While these are valid findings from the RSA-Bench paper, presenting them as definitive laws without acknowledging they are specific observations from a single benchmark risks over-claiming their universality. The text should clarify that these are "observed phenomena in RSA-Bench" rather than general principles of LALM architecture.

These issues are primarily matters of precision in language and scope. The underlying science is sound, but the narrative occasionally stretches beyond the specific data points cited.

Action items.

- **[writing]** The claim that 'offensive research is mature, while defenses are rudimentary' (Conclusion, Sec 5.4) is an over-generalization. The paper cites numerous specific defense mechanisms (ALMGuard, SARSteer, AudioSafe) with quantitative results. The authors should qualify this claim to reflect that defenses are 'fragmented' or 'less standardized' rather than 'rudimentary' to avoid dismissing valid existing work.
- **[writing]** The assertion that LALMs 'infer gender with >92% accuracy' (Sec 5.3.2) based on HearSay is presented as a universal capability of the class of models. This extrapolates from a specific benchmark result to a general property of all LALMs without clarifying if this applies to all architectures or only those tested on that specific dataset. The scope of this claim needs tightening.
- **[writing]** The paper claims a 'Perception-Cognition Gap' and 'Denoising Paradox' (Sec 5.1.1) as established phenomena. While supported by RSA-Bench, presenting these as definitive, named laws of LALM behavior without acknowledging they are specific findings of one benchmark risks over-claiming the universality of these specific failure modes across the entire field.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This survey provides a critical and timely taxonomy of safety and trustworthiness issues in Large Audio Language Models (LALMs), effectively categorizing risks into hallucination, robustness, safety, privacy, fairness, and authentication. The identification of the "asymmetry of offense and defense" (Sec 5.3.1, Sec 6.3) is a significant contribution that highlights the urgent need for defensive research.

However, from a safety and ethics perspective, the manuscript requires minor revisions to address dual-use risks and data provenance concerns:

1. **Dual-Use and Attack Methodology:** Sections 5.3.1 and 5.3.2 provide detailed descriptions of attack vectors (e.g., "WhisperInject," "Multi-AudioJail") and specific success rates (e.g., 21.5% jailbreak success). While essential for benchmarking, the text currently lacks a strong ethical disclaimer. The authors should explicitly frame these findings as defensive tools and warn against the potential misuse of the described methodologies to generate new, unmitigated adversarial attacks. A statement clarifying that these benchmarks are intended solely for improving model robustness is necessary.
2. **Privacy and Consent in Biometric Inference:** The survey cites works like *HearSay* (Sec 5.3.2) which demonstrate high-accuracy inference of gender and identity, and *AGL1K* for geo-localization. Given the sensitive nature of biometric data, the manuscript should

briefly address the ethical implications of these capabilities. Specifically, it should note whether the underlying datasets used in these cited studies were collected with informed consent, and include a warning about the risks of deploying such models in surveillance or non-consensual profiling scenarios.

3. **Safety of Future Proposals:** The "Future Horizons" section (Sec 5.4) proposes "Agent-Based Dynamic Red-Teaming." While innovative, this introduces a new safety vector where autonomous agents could potentially generate harmful content during evaluation. The authors should add a sentence outlining the necessary safety guardrails (e.g., human-in-the-loop, sandboxing) required for such autonomous red-teaming frameworks to ensure they do not become vectors for harm themselves.

The paper is otherwise a strong resource for the community, but these clarifications are vital to ensure the research is applied responsibly.

Action items.

- [**writing**] The manuscript extensively details offensive capabilities (jailbreaks, backdoors, privacy leakage) in Sections 5.3.1 and 5.3.2. To prevent dual-use harm, the authors must explicitly state that the provided attack taxonomies and success rates are for defensive benchmarking only and should not be used to generate new adversarial examples without strict ethical oversight.
- [**writing**] Section 5.3.2 cites 'HearSay' regarding identity inference (>92% accuracy) and 'AGL1K' for geo-localization. The paper must clarify the consent status of the datasets used in these cited works. If the data involves non-consensual biometric inference, the survey should include a dedicated ethical warning about the deployment risks of such models in surveillance contexts.
- [**writing**] The 'Future Outlook' proposes 'Automated Red Teaming agents' (Sec 5.4). The authors should add a brief discussion on the safety protocols required to prevent these autonomous agents from inadvertently generating harmful content or escalating attacks during the evaluation process.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

This survey provides a broad taxonomy of Large Audio Language Models (LALMs) but relies heavily on qualitative summaries of external benchmarks without providing the statistical rigor required to substantiate its central claims regarding performance gaps and security vulnerabilities.

The primary scientific weakness is the absence of quantitative evidence supporting specific comparative claims. For instance, in Section 5.3.1, the paper states that "audio attacks achieve higher success rates (21.5%) than text (17.0%)" based on JALMBench. Without reporting the sample size (N) of the attack attempts, the variance across different models, or statistical significance tests (e.g., p-values, confidence intervals), it is impossible to determine if this 4.5% difference is a robust finding or a result of random noise or specific dataset idiosyncrasies. Similarly, the claim in Section 5.2.1 that ChronosAudio reveals "performance drops exceeding 90%" in long contexts is a dramatic quantitative assertion that lacks necessary context: the baseline metric, the specific model configuration, and the control group (short context) performance are not explicitly detailed in the text, making the magnitude of the "Structural Attention Dilution" effect unverifiable from the manuscript alone.

Furthermore, the discussion of backdoor vulnerabilities in Section 5.3.1 cites that "backdoors triggered by emotion/speaking rate achieve >90% success with only 3% poisoning." This is a critical security claim. The review requires the authors to specify the experimental setup: the total dataset size, the specific poisoning injection method, and the evaluation protocol. Without these details, the claim that such a low poisoning rate yields such high success could be an artifact of a specific, non-generalizable experimental design rather than a fundamental property of LALMs.

As a survey paper, the authors are aggregating results from numerous external studies. However, the scientific evidence presented is currently limited to point estimates without measures of uncertainty or replication details. To strengthen the scientific validity of the review, the authors should either include a meta-analysis of the reported metrics (with variance) or explicitly qualify these claims as preliminary observations from specific, non-replicated studies. The current presentation risks overstating the certainty of these findings.

Action items.

- [**science**] The claim that JALMBench shows audio attacks achieve 21.5% success vs 17.0% for text (Sec 5.3.1) lacks statistical context. Please report the sample size (N), confidence intervals, or p-values to confirm this difference is not due to random variance or dataset bias.
- [**science**] The assertion that ChronosAudio reveals performance drops exceeding 90% in long contexts (Sec 5.2.1) requires clarification on the baseline. Specify the exact metric (e.g., accuracy, F1), the specific model tested, and the control condition (short context) to validate the magnitude of the reported degradation.
- [**science**] The statement that AudioSafe backdoors achieve >90% success with only 3% poisoning (Sec 5.3.1) needs methodological detail. Define the poisoning strategy, the target model architecture, and the evaluation protocol to ensure the result is robust and not an artifact of a specific, non-representative setup.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

This survey provides a comprehensive taxonomy of Large Audio Language Models (LALMs) but lacks statistical rigor in its quantitative claims. As a survey paper, the authors aggregate results from various external benchmarks; however, the presentation of these numbers often omits essential statistical context required to interpret the magnitude and reliability of the findings.

Specifically, in Section 5.1.2, the text states that the best LALM achieves "only 63.19 F1" on the BRACE benchmark. It is unclear if this is a single model's score or an average. If it represents an aggregate, the number of models tested (N) and the dispersion (standard deviation or confidence interval) are missing. Without this, the claim that performance is "only" 63.19 is subjective and statistically unsupported.

In Section 5.3.1, the authors compare attack success rates: "audio attacks achieve higher success rates (21.5%) than text (17.0%)." This 4.5% difference is presented as a definitive finding. The manuscript must specify the statistical test used (e.g., McNemar's test for paired comparisons or a two-proportion z-test) and report the p-value or 95% confidence interval to confirm that this difference is not due to random variation in the test sets.

Furthermore, Section 5.2.2 claims that "answer option permutation can change accuracy by up to 24%." The phrase "up to" suggests a maximum observed effect rather than a central tendency. To be scientifically sound, the authors should report the sample size (N) of the permutation experiments and the distribution of the accuracy drops (e.g., mean ± SD). A single outlier could drive the "up to" figure, potentially misrepresenting the general robustness of the models.

Finally, while Table 1 and Table 2 list numerous benchmarks, the text frequently cites specific accuracy or F1 scores (e.g., 53.60%, 39.35% in Section 5.1.3) without clarifying if these are means across multiple runs or single-point estimates. For a survey synthesizing data from disparate sources, explicitly stating the statistical summary method (mean, median, best-of-N) and providing variance metrics where available is essential for reproducibility and accurate interpretation.

Action items.

- [**science**] Section 5.1.2 cites BRACE achieving 63.19 F1. As a survey, clarify if this is a single-point estimate or an aggregate. If aggregated, report the number of models tested (N) and the variance (SD/CI) to assess the stability of this claim against the 'best model' framing.
- [**science**] Section 5.3.1 states audio attacks achieve 21.5% success vs 17.0% for text. Specify the statistical test used to determine if this 4.5% difference is significant (e.g., McNemar's test for paired data or a z-test for proportions) and report the p-value or confidence interval.
- [**science**] Section 5.2.2 claims answer permutation changes accuracy by 'up to 24%'. Define the baseline (random chance vs. original order) and the sample size (N) of the permutation test. Without N and variance, this 'up to' claim lacks statistical rigor.
- [**writing**] Table 1 and Table 2 list numerous benchmarks with specific metrics (e.g., WER, Accuracy). For any benchmark where the authors summarize performance across multiple

models, ensure the text or table footnotes specify if the reported value is the mean, median, or best-case, and include the standard deviation.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a comprehensive survey of Large Audio Language Models (LALMs) with a strong focus on trustworthiness. However, the writing quality suffers from structural redundancy and occasional syntactic ambiguity that impedes readability.

The most significant issue is the repetition of core content between the Introduction and the main body. Specifically, the definition of the three evaluation pillars (Fidelity, Stability, Alignment) and the description of the "six pillars" of trustworthiness appear verbatim in both the Introduction (e001) and Section 5 (e000). This duplication breaks the logical flow, making the paper feel disjointed. The Introduction should provide a high-level roadmap, while the detailed taxonomy belongs exclusively in Section 5.

Additionally, several sentences suffer from "headline-style" phrasing where benchmark names or paper titles are used as grammatical subjects without clear verbs or context. For instance, in Section 5.1.2, "BRACE [cite] shows the best LALM achieves..." is slightly awkward; it is clearer to state, "The BRACE benchmark reveals that the best LALM achieves..." Similarly, in Section 5.3.1, "Safety under Emotional Variations [cite] reveals..." should be rephrased to attribute the finding to the authors or the study rather than the title itself.

Finally, the text relies heavily on custom LaTeX macros (e.g., `\YearRow`, `\tablogo`) that are not defined in the visible snippets. While this is a technical compilation issue, it obscures the prose structure in the source. The authors should ensure all custom commands are properly defined in the preamble to maintain a clean, readable manuscript. Addressing these flow and clarity issues will significantly enhance the paper's professional presentation.

Action items.

- **[writing]** In Section 5.1.2, the sentence 'BRACE [cite] shows the best LALM achieves only 63.19 F1 on reference-free caption alignment' lacks a clear subject-verb connection to the benchmark name. Rephrase to explicitly state that the benchmark 'BRACE' evaluates this metric, e.g., 'On the BRACE benchmark, the best LALM achieves...'
- **[writing]** The Introduction (e001) and Section 5 (e000) contain nearly identical definitions of the three evaluation pillars (Fidelity, Stability, Alignment) and the same figure/table references. This redundancy disrupts the narrative flow. Consolidate the detailed taxonomy into Section 5 and keep the Introduction summary high-level.
- **[writing]** In Section 5.3.1, the phrase 'Safety under Emotional Variations [cite] reveals Emotional Hijacking...' uses a paper title as a proper noun subject. Cite the specific author

or model name if available, or rephrase to ‘Research on safety under emotional variations [cite] reveals...’

- **[writing]** Table 1 (e001) and Table 2 (e002) both use custom macros like \YearRow and \tablogo which are not defined in the provided snippets. While this is a compilation issue, the text flow is interrupted by these undefined commands. Ensure all custom commands are defined in the preamble or replace with standard LaTeX to ensure readability of the source.